

Laboratory Guide 4 (Finance): Quantum Portfolio Optimization

Finance & Industry Applications of Variational Quantum Algorithms

Course: Quantum Computing Foundation

Contents

1	Introduction and Objectives	2
2	Part A: Classical Portfolio Theory	2
2.1	Mean-Variance Optimization (Markowitz, 1952)	2
2.2	The Combinatorial Version and its Hardness	3
2.3	Asset Universe: AAPL, MSFT, JPM, XOM	3
3	Part B: QUBO Formulation	4
3.1	From Constrained to Unconstrained	4
3.2	Classical Brute-Force Baseline	5
4	Part C: Quantum Hamiltonian and QAOA Circuit	6
4.1	QUBO $\rightarrow$ Ising Hamiltonian	6
4.2	Explicit Hamiltonian Terms ($n = 4, \lambda = 1$)	6
4.3	QAOA Circuit Structure ($p = 1$)	7
5	Part D: Result Interpretation	8
5.1	Reading the Output Distribution	8
5.2	Approximation Quality and Business Risk	8
5.3	Sensitivity Analysis: Risk Aversion λ	8
6	Part E: IBM Composer Exercise	9
7	Part F: Qiskit Python Implementation	9
8	Part G: Scaling and Outlook	10
8.1	Scaling to Larger Portfolios	10
8.2	Limitations and Open Questions	10
9	Part H: Assessment	11

1 Introduction and Objectives

This laboratory applies a quantum optimization algorithm to a real-world industrial problem: **portfolio optimization in quantitative finance**. Everything it needs beyond Labs 1–2 is explained in this guide. If you did Lab 4, you have met the same algorithm on a small graph problem (MaxCut), and the optional boxes point out the connections.

The investment choice is written as a Quadratic Unconstrained Binary Optimization (QUBO) problem, rewritten as an operator on qubits (an Ising Hamiltonian), and searched with a short quantum circuit, the Quantum Approximate Optimization Algorithm (QAOA). The novelty is threefold: (i) the QUBO is built from (stylised) asset data, (ii) interpreting the output requires both a technical and a business lens, and (iii) the problem has a natural classical solution that allows direct benchmarking.

Prerequisites:

- Labs 1–2 (Lab 4 helpful, not required)
- Basic probability and statistics: mean, variance, covariance
- No prior knowledge of quantitative finance is assumed

Learning Objectives:

1. Model a combinatorial portfolio selection problem as a QUBO.
2. Map the QUBO to a quantum Hamiltonian and build the QAOA circuit.
3. Run QAOA on a 4-asset universe (AAPL, MSFT, JPM, XOM) and verify the result against brute-force.
4. Interpret results from both a business and a technical perspective.
5. Understand how the circuit and the classical search space grow with n , and what is (not yet) known about quantum advantage.

Why Quantum for Portfolio Optimization?

For $n = 4$, $K = 2$ there are only 6 portfolios, and a laptop checks them all instantly. For $n = 50$, $K = 10$ there are $\binom{50}{10} \approx 10^{10}$: checking every one becomes impractical, and real desks use classical heuristics instead. QAOA's circuit grows only as $\mathcal{O}(n^2)$ gates per layer, which is why it is studied for this problem. Whether it can beat the best classical heuristics is an open research question. This lab shows what it does and does not deliver today.

2 Part A: Classical Portfolio Theory

2.1 Mean-Variance Optimization (Markowitz, 1952)

Let $x \in \mathbb{R}^n$ be a portfolio weight vector with $\sum_i x_i = 1$ (continuous). The mean-variance problem is:

$$\min_x \lambda x^T \Sigma x - \mu^T x, \quad (1)$$

where $\Sigma \in \mathbb{R}^{n \times n}$ is the asset return *covariance matrix*, $\mu \in \mathbb{R}^n$ is the vector of *expected returns*, and $\lambda > 0$ is the *risk-aversion* parameter. The first term penalises risk (portfolio variance), the second rewards expected return.

Risk-Aversion Parameter λ

- $\lambda \rightarrow 0$: maximise returns regardless of risk (aggressive investor).
- $\lambda \rightarrow \infty$: minimise variance regardless of returns (defensive investor, cash-like).
- In this lab we explore $\lambda \in [0.5, 5]$.

For this lab we use $\lambda = 1$ as the baseline.

2.2 The Combinatorial Version and its Hardness

In practice, portfolio managers must select *exactly* K assets from a universe of n candidates.

Declared simplification: a portfolio here is a *selection* — each chosen asset gets the same amount of money; choosing weights continuously (the full Markowitz problem) is not modelled. The binary version of equation (1) is:

$$\min_{x \in \{0,1\}^n} \lambda x^T \Sigma x - \boldsymbol{\mu}^T x, \quad \text{subject to} \quad \mathbf{1}^T x = K. \quad (2)$$

For $n = 4$, $K = 2$ there are only $\binom{4}{2} = 6$ candidates — trivially enumerable. For $n = 50$, $K = 10$ there are $\binom{50}{10} \approx 10^{10}$ candidates. For $n = 500$, $K = 50$: $\approx 2.3 \times 10^{69}$ — far beyond brute-force. The continuous relaxation (equation (1)) gives the *efficient frontier* but loses the discreteness that real portfolios require.

Efficient frontier. A portfolio is *efficient* if no other portfolio offers at least the same expected return for less risk, or a higher expected return for no more risk. The efficient ones, drawn in the risk–return plane, form the *efficient frontier*; every other point is *dominated*. Among our six 2-asset portfolios (equal weights), four are efficient (MSFT + JPM, AAPL + JPM, AAPL + XOM, AAPL + MSFT); MSFT + XOM and JPM + XOM are dominated. Moving λ walks the optimum along this frontier.

2.3 Asset Universe: AAPL, MSFT, JPM, XOM

We work with the four assets in Table 1. The covariance matrix Σ and expected returns $\boldsymbol{\mu}$ are stylised teaching values (annualised, rounded), not an estimate from a specific price history.

Table 1: Asset data used in this laboratory (annualised, stylised teaching values).

Index	Ticker	Sector	μ_i	$\sigma_i = \sqrt{\Sigma_{ii}}$
0	AAPL	Technology	0.340	0.290
1	MSFT	Technology	0.270	0.270
2	JPM	Financials	0.150	0.240
3	XOM	Energy	0.220	0.350

The annualised covariance matrix (stylised):

$$\Sigma = \begin{pmatrix} 0.0841 & 0.0588 & 0.0313 & 0.0203 \\ 0.0588 & 0.0729 & 0.0272 & 0.0170 \\ 0.0313 & 0.0272 & 0.0576 & 0.0294 \\ 0.0203 & 0.0170 & 0.0294 & 0.1225 \end{pmatrix}. \quad (3)$$

The diagonal entries $\Sigma_{ii} = \sigma_i^2$ are the individual asset variances; the off-diagonal entries capture return correlations. AAPL–MSFT have the highest co-movement (both tech-sector), while XOM (energy) is weakly correlated with the others.

3 Part B: QUBO Formulation

3.1 From Constrained to Unconstrained

We convert the equality constraint $\sum_i x_i = K$ into a quadratic penalty added to the objective:

$$\min_{x \in \{0,1\}^n} \underbrace{\lambda x^T \Sigma x - \boldsymbol{\mu}^T x}_{\text{cost}} + \underbrace{\lambda_{\text{pen}} \left(\sum_{i=1}^n x_i - K \right)^2}_{\text{budget penalty}}, \quad (4)$$

where λ_{pen} must be large enough that any infeasible solution (wrong number of assets) is always worse than any feasible one. Since every infeasible x pays a penalty of at least λ_{pen} , a sufficient choice is $\lambda_{\text{pen}} > \max_{x \text{ feasible}} C(x) - \min_{x \text{ infeasible}} C(x)$; with $C(x) = \lambda x^T \Sigma x - \boldsymbol{\mu}^T x$ the cost term; for our data this bound is about 0.23 at $\lambda = 1$ and at most 0.83 for $\lambda \leq 5$, so $\lambda_{\text{pen}} = 3$ is ample.

Expanding and collecting, equation (4) takes the standard QUBO form.

QUBO Derivation — step by step

Since $x_i \in \{0,1\}$, we have $x_i^2 = x_i$. Expand the budget penalty (dropping the constant $\lambda_{\text{pen}} K^2$):

$$\lambda_{\text{pen}} \left(\sum_i x_i - K \right)^2 \rightarrow \lambda_{\text{pen}} (1 - 2K) \sum_i x_i + 2\lambda_{\text{pen}} \sum_{i < j} x_i x_j.$$

Adding the financial objective and grouping:

- **Diagonal** ($i = j$): coefficient of x_i is $\lambda \Sigma_{ii} - \mu_i + \lambda_{\text{pen}} (1 - 2K)$.
- **Off-diagonal** ($i < j$, one occurrence): coefficient of $x_i x_j$ is $2\lambda \Sigma_{ij} + 2\lambda_{\text{pen}}$.

Convention note: because the full product $x^T Q x = \sum_{i,j} Q_{ij} x_i x_j$ counts each unordered pair twice, the symmetric matrix entry Q_{ij} ($i \neq j$) in equation (6) carries *half* of the $i < j$ coefficient above — the risk part and the penalty part alike — giving $Q_{ij} = \lambda \Sigma_{ij} + \lambda_{\text{pen}}$. With this scaling,

$$x^T Q x = C(x) + \lambda_{\text{pen}} \left(\sum_i x_i - K \right)^2 - \lambda_{\text{pen}} K^2.$$

On a feasible portfolio the penalty vanishes, so $x^T Q x = C(x) - \lambda_{\text{pen}} K^2$: a constant shift, identical for every feasible x , which does not affect the ranking. Every infeasible portfolio pays a strictly positive penalty — that is what enforces the budget constraint.

$$\min_{x \in \{0,1\}^n} x^T Q x, \quad (5)$$

with:

$$Q_{ij} = \begin{cases} \lambda \Sigma_{ii} - \mu_i + \lambda_{\text{pen}} (1 - 2K) & i = j \\ \lambda \Sigma_{ij} + \lambda_{\text{pen}} & i \neq j. \end{cases} \quad (6)$$

QUBO Matrix for $n = 4$, $K = 2$, $\lambda = 1$, $\lambda_{\text{pen}} = 3$

Substituting equation (3) and $\mu = (0.34, 0.27, 0.15, 0.22)$ into equation (6) with $\lambda_{\text{pen}}(1 - 2K) = 3(1 - 4) = -9$:

$$Q = \lambda \Sigma - \text{diag}(\mu) + \lambda_{\text{pen}} \begin{pmatrix} 1-2K & 1 & 1 & 1 \\ 1 & 1-2K & 1 & 1 \\ 1 & 1 & 1-2K & 1 \\ 1 & 1 & 1 & 1-2K \end{pmatrix}$$

$$Q = \begin{pmatrix} -9.2559 & 3.0588 & 3.0313 & 3.0203 \\ 3.0588 & -9.1971 & 3.0272 & 3.0170 \\ 3.0313 & 3.0272 & -9.0924 & 3.0294 \\ 3.0203 & 3.0170 & 3.0294 & -9.0975 \end{pmatrix}.$$

The positive off-diagonal terms ($\approx \lambda_{\text{pen}}$) reflect the budget penalty, which charges every extra pair of selected assets; the large negative diagonal terms ($\approx \lambda_{\text{pen}}(1 - 2K) = -9$) come from the linear part of the same penalty, and the small remainder $\lambda \Sigma_{ii} - \mu_i$ on the diagonal is the finance (own risk minus return).

3.2 Classical Brute-Force Baseline

For $n = 4$, $K = 2$ we can enumerate all $\binom{4}{2} = 6$ portfolios exactly. Table 2 shows the financial cost $C(x) = \lambda x^T \Sigma x - \mu^T x$ for each feasible portfolio. (All feasible portfolios satisfy the budget constraint, so the penalty term is zero and $C(x)$ equals the QUBO cost up to the constant $\lambda_{\text{pen}} K^2 = 12$; only differences matter for ranking.)

Table 2: Brute-force enumeration of all 2-asset portfolios ($\lambda = 1$). Portfolio vector $x = [x_0, x_1, x_2, x_3]$ shown in AAPL/MSFT/JPM/XOM order.

Portfolio vector x	Assets selected	Financial cost $C(x)$	Rank
1100	AAPL + MSFT	-0.335	1 (optimal)
1001	AAPL + XOM	-0.313	2
1010	AAPL + JPM	-0.286	3
0101	MSFT + XOM	-0.261	4
0110	MSFT + JPM	-0.235	5
0011	JPM + XOM	-0.131	6

Financial Interpretation

Why AAPL + MSFT? Despite having the highest inter-asset covariance (both technology sector), AAPL and MSFT also have the highest expected returns (0.34 and 0.27 respectively). For $\lambda = 1$ the return advantage outweighs the diversification cost. As λ increases (more risk-averse investor), the optimal portfolio shifts to AAPL + JPM (around $\lambda = 2$) and eventually to MSFT + JPM (by $\lambda = 5$). Both shifts bring in JPM, which has the smallest individual variance ($\Sigma_{\text{JPM}, \text{JPM}} = 0.0576$). Note that XOM never enters the optimal portfolio, despite having the smallest covariances with the other assets ($\Sigma_{\text{MSFT}, \text{XOM}} = 0.0170$ is the smallest off-diagonal entry of all): portfolio variance $x^T \Sigma x$ includes the diagonal, and XOM's own variance (0.1225, i.e. $\sigma = 35\%$) is by far the largest. Low correlation cannot rescue an asset that is individually too volatile.

4 Part C: Quantum Hamiltonian and QAOA Circuit

4.1 QUBO $\rightarrow$ Ising Hamiltonian

Substituting $x_i = (1 - Z_i)/2$ into equation (5). Note that the eigenvalue of Z_i on state $|q_i\rangle$ is $+1$ if $q_i = 0$ and -1 if $q_i = 1$, so $x_i = (1 - Z_i)/2$ maps $x_i = 0 \leftrightarrow Z_i = +1$ and $x_i = 1 \leftrightarrow Z_i = -1$ consistently.

$$\begin{aligned} x^T Q x &= \sum_{i,j} Q_{ij} x_i x_j = \sum_{i,j} Q_{ij} \frac{(1 - Z_i)(1 - Z_j)}{4} \\ &= \frac{1}{4} \sum_{i,j} Q_{ij} (I - Z_i - Z_j + Z_i Z_j) \\ &\equiv H_{\text{portfolio}} + \text{const.} \end{aligned} \quad (7)$$

To collect terms, use: (a) $Z_i^2 = I$ (constant, absorbed into “const”); (b) the $-Z_i$ and $-Z_j$ pieces with symmetric Q give $-\frac{1}{2} \sum_j Q_{ij}$ as coefficient of Z_i ; (c) cross terms $Z_i Z_j$ with $i \neq j$ give coefficient $Q_{ij}/2$ after accounting for the $1/4$ prefactor and the factor 2 from $\sum_{i \neq j} = 2 \sum_{i < j}$ (since Q is symmetric). Collecting terms, the cost Hamiltonian is:

$$H_{\text{portfolio}} = \sum_{i < j} \frac{Q_{ij}}{2} Z_i Z_j + \sum_i \left[-\frac{1}{2} \sum_j Q_{ij} \right] Z_i + \text{const.} \quad (8)$$

A sum of Z and ZZ terms that plays the role of an energy is called an **Ising Hamiltonian** (after a physics model of magnets). The cheapest portfolio is its lowest-energy basis state, the **ground state**: minimising cost and finding the ground state are the same task. Because the substitution drops a constant, energies of $H_{\text{portfolio}}$ are the QUBO costs shifted by a fixed amount (here 9.2294): the best cost -12.3354 corresponds to the ground-state energy -3.1060 .

Optional: connection to Lab 4

The structure of $H_{\text{portfolio}}$ is a sum of $Z_i Z_j$ and Z_i Pauli operators. MaxCut in Lab 4 has only $Z_i Z_j$ terms, all with the same coefficient; the circuit is built from the same blocks:

- $e^{-i\gamma J_{ij} Z_i Z_j}$: CNOT $- R_z(2\gamma J_{ij})$ $-$ CNOT
- $e^{-i\gamma h_i Z_i}$: $R_z(2\gamma h_i)$ directly
- $e^{-i\beta H_B}$: $R_x(2\beta)$ on all qubits

Here the coefficients $J_{ij} = Q_{ij}/2$ and h_i come from the financial data, and the single- Z terms are new.

4.2 Explicit Hamiltonian Terms ($n = 4$, $\lambda = 1$)

Applying equation (8) to our Q matrix yields the following Pauli terms and constant (the same numbers the companion notebook prints):

Pauli term	Coefficient	Origin
$Z_0 Z_1$	+1.5294	$Q_{01}/2$ (AAPL–MSFT covariance + penalty)
$Z_0 Z_2$	+1.5156	$Q_{02}/2$ (AAPL–JPM covariance + penalty)
$Z_0 Z_3$	+1.5102	$Q_{03}/2$ (AAPL–XOM covariance + penalty)
$Z_1 Z_2$	+1.5136	$Q_{12}/2$ (MSFT–JPM covariance + penalty)
$Z_1 Z_3$	+1.5085	$Q_{13}/2$ (MSFT–XOM covariance + penalty)
$Z_2 Z_3$	+1.5147	$Q_{23}/2$ (JPM–XOM covariance + penalty)
Z_0	+0.0728	$-\frac{1}{2} \sum_j Q_{0j}$ (AAPL)
Z_1	+0.0471	$-\frac{1}{2} \sum_j Q_{1j}$ (MSFT)
Z_2	+0.0022	$-\frac{1}{2} \sum_j Q_{2j}$ (JPM)
Z_3	+0.0154	$-\frac{1}{2} \sum_j Q_{3j}$ (XOM)
Constant	−9.2294	dropped: $x^T Q x = H_{\text{portfolio}} - 9.2294$

Ground-state energy: $-3.1060 = -12.3354 + 9.2294$ (AAPL + MSFT).

QAOA in one box

QAOA — Quantum Approximate Optimization Algorithm (developed in full in Lab 4; everything needed here is below).

1. **Start:** a Hadamard gate on every qubit gives all $2^4 = 16$ bit strings the same probability, $1/16$.
2. **Cost layer** $e^{-i\gamma H_{\text{portfolio}}}$: gives every bit string a phase proportional to its cost; γ is an angle we choose. A phase alone changes no measurement probability. Each $Z_i Z_j$ term is built as CNOT– R_z –CNOT: the first CNOT writes “do bits i and j differ?” onto qubit j , the R_z applies a phase that depends on that answer, and the second CNOT undoes the first. Each single Z_i term is one R_z on qubit i .
3. **Mixer layer:** an $R_x(2\beta)$ rotation on every qubit makes bit strings that differ by one flip interfere, and this turns phase differences into probability differences.
4. **Repeat** cost + mixer p times (here $p = 1$ or 2): $2p$ angles in total.
5. **Tune:** run the circuit many times, average the cost of the outcomes (this average is written $\langle H \rangle$ and called the *expectation value*), and let an ordinary classical optimizer (COBYLA, a standard derivative-free routine) change the angles to lower that average. Nothing in this step is quantum.

The hope is that cheap bit strings end up with high probability. Whether that happens here is what Part D reports.

4.3 QAOA Circuit Structure ($p = 1$)

The $p = 1$ QAOA circuit starts with a Hadamard layer ($H^{\otimes 4}$: the Hadamard gate H on every qubit, not the Hamiltonian $H_{\text{portfolio}}$), then alternates one cost unitary and one mixer:

$$|\gamma, \beta\rangle = e^{-i\beta \sum_i X_i} e^{-i\gamma H_{\text{portfolio}}} H^{\otimes 4} |0000\rangle. \quad (9)$$

Per cost layer: one CNOT– R_z –CNOT block for each of the $\binom{4}{2} = 6$ $Z_i Z_j$ terms (12 CNOTs, 6 R_z), plus one R_z for each of the 4 Z_i terms; the mixer adds one R_x per qubit. The two-qubit part grows as $\binom{n}{2} = \mathcal{O}(n^2)$.

5 Part D: Result Interpretation

5.1 Reading the Output Distribution

After QAOA optimisation, we sample the circuit M times and obtain a histogram over $2^n = 16$ bitstrings. The *technical* interpretation: the cheapest portfolio, AAPL + MSFT, is the state $|1100\rangle$ (lab order $|q_0q_1q_2q_3\rangle$, convention A1; Qiskit prints 0011). Ideally it would carry the highest probability; the next subsection shows that here it does not.

Technical → Business Translation

Bitstring	Business meaning
$ 1100\rangle$	Invest in AAPL and MSFT
$ 1001\rangle$	Invest in AAPL and XOM
$ 0011\rangle$	Invest in JPM and XOM
$ 0000\rangle$	Infeasible (no assets)
$ 1111\rangle$	Infeasible (all assets, $K \neq 2$)

Infeasible states have high energy due to the penalty term: at $p = 1$ they still carry about 26% of the probability, at $p = 2$ almost none.

5.2 Approximation Quality and Business Risk

Define the *success probability* $P^* = P(|1100\rangle)$: the probability the circuit returns the exact optimal portfolio. With the angles optimised in a noiseless simulation (single- Z terms included):

- $p = 1$: about 74% of the probability sits on the six feasible portfolios, each $\approx 12\%$, so $P^* \approx 12\%$; about 26% is still infeasible.
- $p = 2$: essentially 100% is feasible, spread almost evenly ($\approx 16.7\%$ each, $P^* \approx 16.5\%$).

QAOA enforces the budget constraint, but it does not pick the winner: the six feasible costs differ by at most 0.21 and the best two by only 0.023, far below what one or two layers can resolve. The quantum step supplies a *shortlist*; the exact cost ranks it. For production use, we would:

1. Sample the circuit M times (e.g. $M = 1000$).
2. Keep the *feasible* bitstrings that appear (the shortlist).
3. Rank them by their exact cost $C(x)$, not by how often they were measured.

Increasing p raises the feasible fraction but also increases circuit depth and susceptibility to hardware noise.

5.3 Sensitivity Analysis: Risk Aversion λ

How λ Shifts the Optimal Portfolio

At $\lambda = 0.5$ (return-focused): **AAPL + MSFT** optimal.
 At $\lambda = 1.0$ (balanced): **AAPL + MSFT** optimal.
 At $\lambda = 2.0$ (risk-averse): **AAPL + JPM** takes over.
 At $\lambda = 5.0$ (very defensive): **MSFT + JPM** (lowest-variance pair) wins.
 The quantum circuit must be re-optimised for each λ because the Hamiltonian $H_{\text{portfolio}}(\lambda)$ changes. The optimal QAOA parameters (γ^*, β^*) also shift.

6 Part E: IBM Composer Exercise

Platform: IBM Quantum Composer — <https://quantum.ibm.com/composer>

Student Activity

Build one full QAOA layer ($p = 1$) by hand. The six $Z_i Z_j$ coefficients are equal to within 1.4% (1.5085–1.5294, see the Hamiltonian table in Part C), so one angle serves them all. The four Z_i terms are at most 5% of a ZZ term and are omitted here: this is a **declared simplification**, and it changes the feasible fraction by less than 0.5 percentage points.

1. Put H on $q_0, \dots, q_3$.
2. For each of the 6 pairs (q_i, q_j) , $i < j$, add $\text{CNOT}(q_i, q_j) - R_z(0.50)$ on $q_j - \text{CNOT}(q_i, q_j)$. These 18 gates are the cost layer with $\gamma \approx 0.16$.
3. Add $R_x(-0.56)$ on every qubit. This is the mixer with $\beta \approx -0.28$. If the Composer accepts only positive angles, use 5.72.
4. Open the probabilities view. The six bit strings with exactly two 1s now carry $\approx 74\%$ in total ($\approx 12\%$ each).

Compare with two variants:

- **No cost layer** (only H , then R_x): $37.5\% = 6/16$. The mixer alone does nothing useful.
- $R_x(+0.56)$ instead of $R_x(-0.56)$: $\approx 18\%$. The same gates with the opposite sign push probability *away* from the feasible portfolios.

Observation: the angles are the whole algorithm, and that is why a classical optimizer has to tune them.

7 Part F: Qiskit Python Implementation

The full implementation is in the companion notebook. Three exercises are provided:

1. **QUBO Construction:** Build Q from the asset data, enumerate all $\binom{4}{2}$ portfolios classically, identify the optimum.
2. **QAOA Optimisation:** Run QAOA with $p \in \{1, 2\}$, compare the measured distribution against brute-force.
3. **Business Analysis:** Plot the efficient frontier, rank the feasible portfolios in the QAOA sample by exact cost, and compute the risk-return profiles of the top three.

Key Code Structure

```

1 # Core objects: same structure as Lab 4, different Hamiltonian
2 def build_portfolio_hamiltonian(Q: np.ndarray) -> SparsePauliOp:
3     n = Q.shape[0]
4     terms = []
5     for i in range(n):
6         for j in range(i+1, n):
7             # zz_str, z_str: Pauli-label helpers, see notebook
8             coeff = Q[i,j] / 2 # ZiZj coefficient
9             terms.append((zz_str(i,j,n), coeff))
10            z_coeff = -sum(Q[i,:]) / 2 # Zi coefficient
11            terms.append((z_str(i,n), z_coeff))
12        return SparsePauliOp.from_list(terms).simplify()
13
14 # QAOA circuit: H layer, then per layer one CNOT-Rz-CNOT per ZZ term,
15 # one Rz per Z term, Rx mixer (see notebook)

```

8 Part G: Scaling and Outlook

8.1 Scaling to Larger Portfolios

Table 3: Portfolio counts and QAOA circuit size (n assets, K selected; one qubit per asset).

n	K	Portfolios $\binom{n}{K}$	ZZ terms $\binom{n}{2}$	CNOTs per cost layer
4	2	6	6	12
6	3	20	15	30
10	5	252	45	90
20	10	184,756	190	380
50	10	$\approx 1.0 \times 10^{10}$	1,225	2,450
50	25	$\approx 1.3 \times 10^{14}$	1,225	2,450

The QAOA circuit grows as $\mathcal{O}(n^2)$ gates (from the $\binom{n}{2}$ ZZ interaction terms), while the classical search space grows as $\binom{n}{K} \sim n^K/K!$. For $n = 20$, $K = 5$: classical brute-force checks $\binom{20}{5} = 15,504 \approx 1.6 \times 10^4$ portfolios (fast classically); for $n = 50$, $K = 10$: $\approx 1.0 \times 10^{10}$ (impractical to enumerate). The QAOA circuit for $n = 50$ needs $\binom{50}{2} = 1225$ CNOT- R_z -CNOT blocks, i.e. 2450 CNOTs per layer: small in gate count, but see the noise estimate below.

8.2 Limitations and Open Questions

- **Barren plateaus:** For $n \gtrsim 50$, the QAOA landscape may be exponentially flat, requiring problem-informed initial parameters.
- **Hardware noise:** The 2450 CX gates of one layer at $n = 50$, with $\sim 0.5\%$ error each, give an overall fidelity of $0.995^{2450} \approx 5 \times 10^{-6}$ — noise dominates.
- **Quantum advantage:** Not yet demonstrated for portfolio optimization. Active research area (JPMorgan, Goldman Sachs, IBM Research).
- **Beyond QUBO:** Transaction costs, lot sizes, turnover constraints are not naturally quadratic — require slack variables or penalty engineering.

9 Part H: Assessment

Q1. QUBO Formulation

Why is the budget constraint $\sum_i x_i = K$ added as a penalty term $\lambda_{\text{pen}}(\sum_i x_i - K)^2$ rather than enforced directly in the quantum circuit?

- A) Because quantum circuits cannot measure the number of 1s in a bitstring.
- B) Because the QAOA Hamiltonian must be a sum of Pauli operators (diagonal in the Z -basis), and a soft penalty converts the hard constraint into such a form.
- C) Because the penalty term is always zero for feasible solutions, making it costless.

Q2. Hamiltonian Terms

In the portfolio Hamiltonian $H_{\text{portfolio}}$, what is the physical origin of the $Z_i Z_j$ terms?

- A) The expected returns μ_i of the individual assets.
- B) The covariance Σ_{ij} between assets i and j , plus the budget penalty terms that couple all pairs of qubits.
- C) The CNOT gates in the QAOA mixer.

Q3. Result Interpretation

At $p = 2$ the QAOA circuit for the 4-asset problem returns each of the six feasible bit strings with probability ≈ 16 –17% and almost nothing else. What should the analyst recommend?

- A) The most frequent bit string in the sample.
- B) Among the feasible bit strings, the one with the lowest exact cost $C(x) = \text{AAPL} + \text{MSFT}$ — with the next-lowest ($\text{AAPL} + \text{XOM}$) as the alternative.
- C) Re-run with more shots until one bit string exceeds 99%.

Q4. Scaling

For $n = 50$ assets and $K = 10$, the classical brute-force requires $\binom{50}{10} \approx 10^{10}$ cost evaluations. How many CNOT gates does one QAOA layer need for $n = 50$?

- A) $n = 50$
- B) $n^2 = 2500$
- C) $2\binom{n}{2} = 2450$ (one CNOT pair per $Z_i Z_j$ term)

Solutions: Q1: B (the Ising Hamiltonian is $\sum_i h_i Z_i + \sum_{ij} J_{ij} Z_i Z_j$; the penalty converts the constraint into this Pauli-operator form); Q2: B (the Q_{ij} for $i \neq j$ contains $\lambda \Sigma_{ij}$ from the risk term plus λ_{pen} from the penalty, both contributing to the $Z_i Z_j$ coefficient after the substitution $x_i = (1 - Z_i)/2$); Q3: B (shot counts differ here only by sampling noise; QAOA supplied the feasible shortlist, and the exact cost ranks it); Q4: C ($\binom{50}{2} = 1225$ $Z_i Z_j$ terms, each built as CNOT- R_z -CNOT, give $2 \times 1225 = 2450$ CNOTs per layer, plus $1225 + 50$ single-qubit R_z and 50 R_x ; total $\mathcal{O}(n^2)$).